



**SCHOOL OF ENGINEERING AND NATURAL
SCIENCES**

EEE

DMCD

PROJECT: 28-Bit CSKA

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1. Introduction

This project focuses on the design, simulation, and physical layout of a 28-bit adder using Cadence Virtuoso. A hierarchical design methodology is adopted, where the adder is constructed from smaller reusable modules. The architecture is optimized to minimize overall propagation delay while preserving correct arithmetic operation and signal integrity.

2. Design Architecture

The proposed 28-bit adder is implemented by cascading seven identical 4-bit adder blocks. Each block is realized using a Carry-Skip Adder (CSKA) structure to improve performance compared to a conventional ripple carry approach. These blocks are connected sequentially through their carry signals, enabling efficient carry propagation across the entire 28-bit word length.

3. Cells Used in the Design

The hierarchical architecture of the 28-bit adder is composed of seven identical 4-bit Ripple Carry Adder (RCA4) modules. Each RCA4 block internally consists of full adder cells responsible for performing basic sum and carry computations. In total, 28 full adders are used to implement the complete 28-bit addition.

Additional logic components, including XOR gates, AND gates, and multiplexers (MUX), are integrated within each block to generate sum outputs and control carry propagation. These gates support propagate and skip logic, allowing faster carry transmission across blocks when possible. This modular and structured design approach simplifies implementation, verification, and layout while ensuring accurate and efficient carry handling throughout the adder.

Cell / Block Type	Description	Quantity
Full Adder (FA)	Performs 1-bit addition within each RCA4 block	28
RCA4 Block	4-bit ripple carry adder module	7
XOR Gate	Used for sum generation and propagate logic	4

Cell / Block Type	Description	Quantity
AND Gate	Used for carry and propagate logic	3
Multiplexer (MUX)	Controls carry selection and skip operation	1

Table 1: Cells Used

Design:

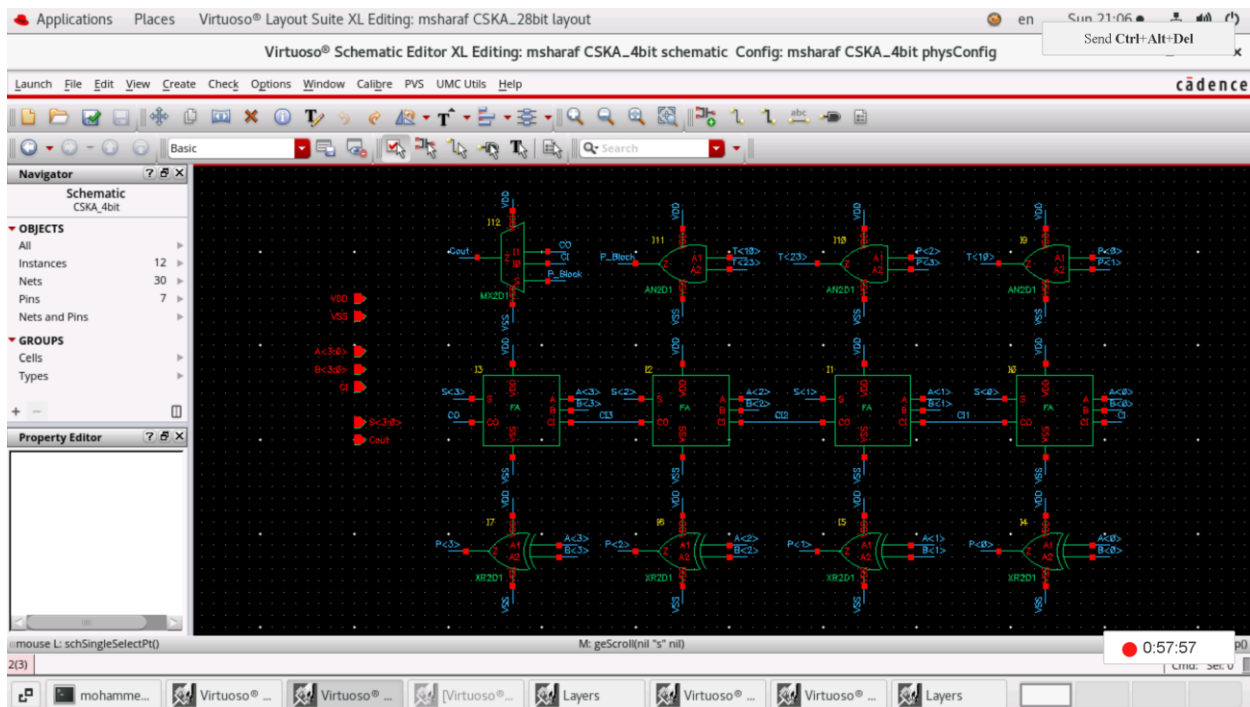


Fig.1 CSKA_4bit Schematic

This schematic represents a 4-bit Carry-Skip Adder (CSKA) designed in Cadence Virtuoso. It is composed of four full adder cells connected in a ripple configuration to generate the sum outputs. XOR gates are used to produce propagate signals for each bit, while AND gates combine these signals to determine whether the carry can bypass the block. A multiplexer then selects between the ripple carry output and the input carry based on the propagate condition. This structure enables faster carry propagation compared to a conventional ripple carry adder while maintaining correct 4-bit addition functionality.

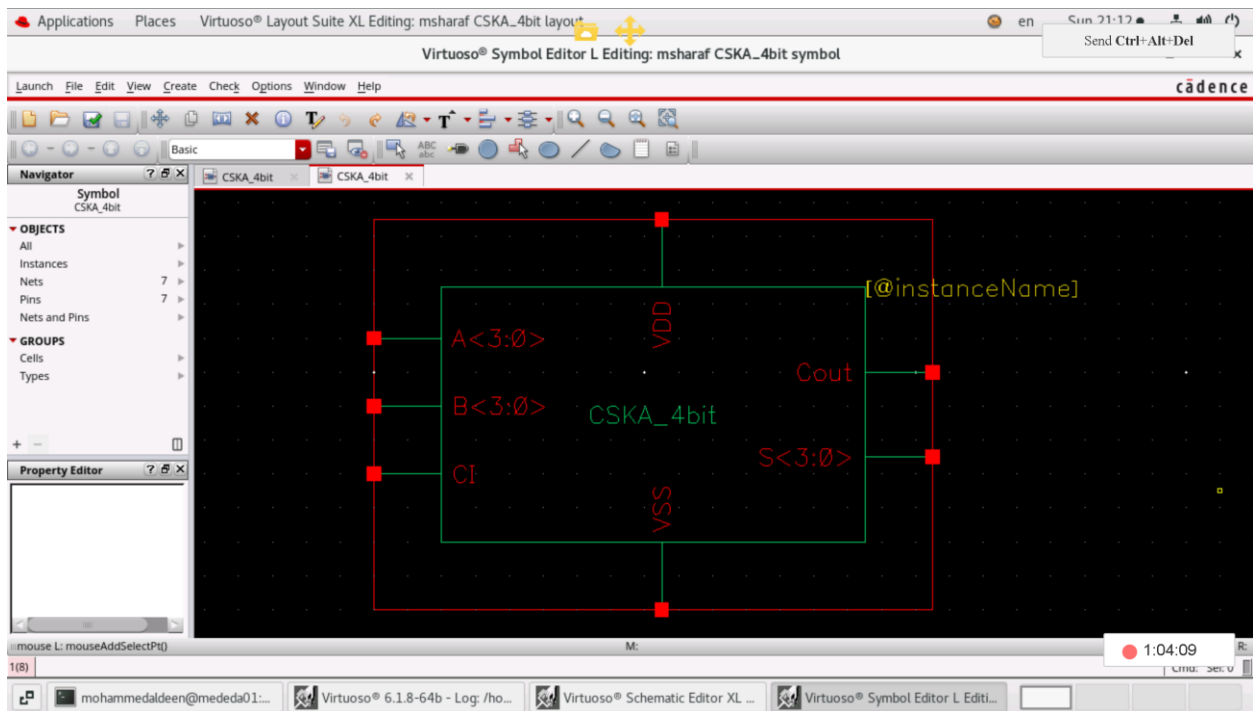


Fig.2 CSKA_4bit Symbol

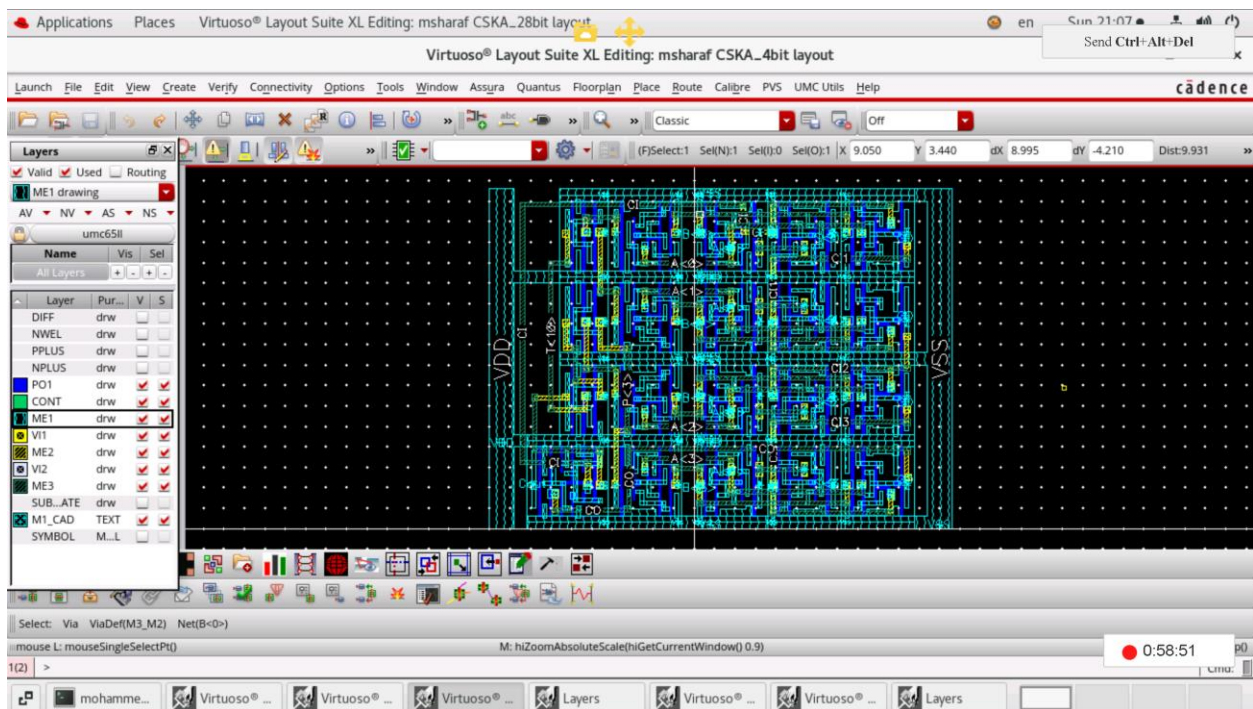


Fig.3 CSKA_4bit Layout

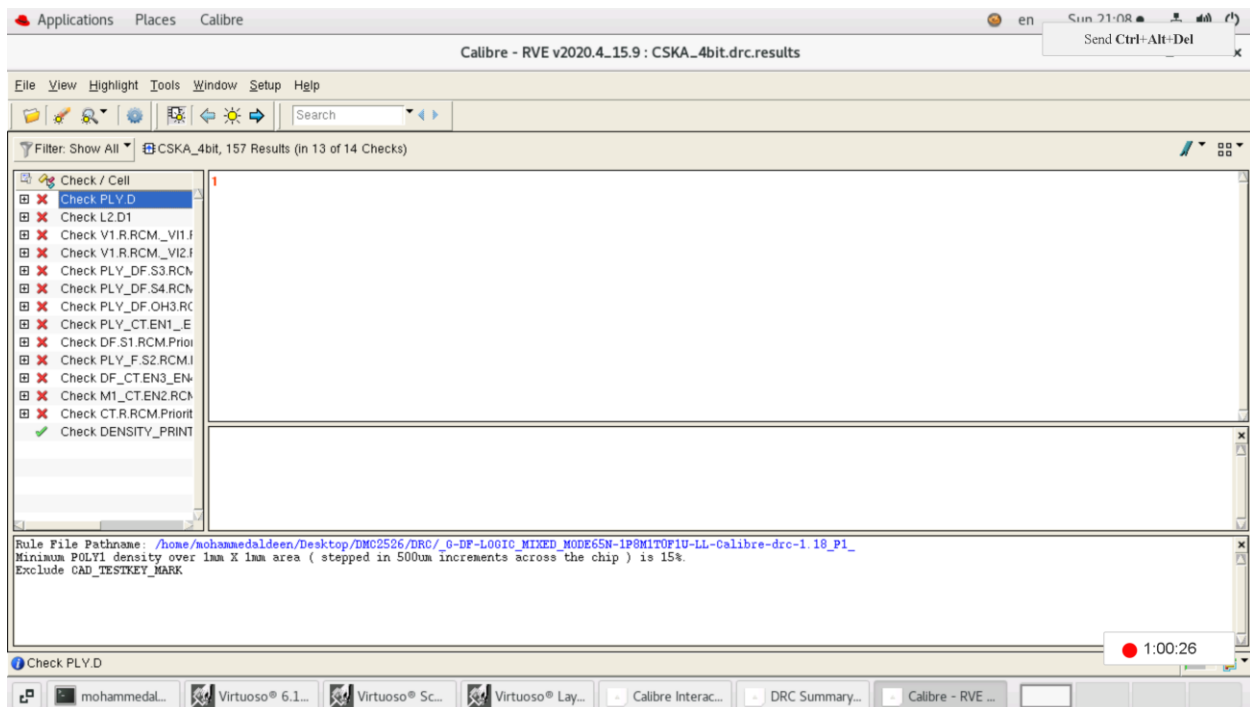


Fig.4 CSKA_4bit DRC

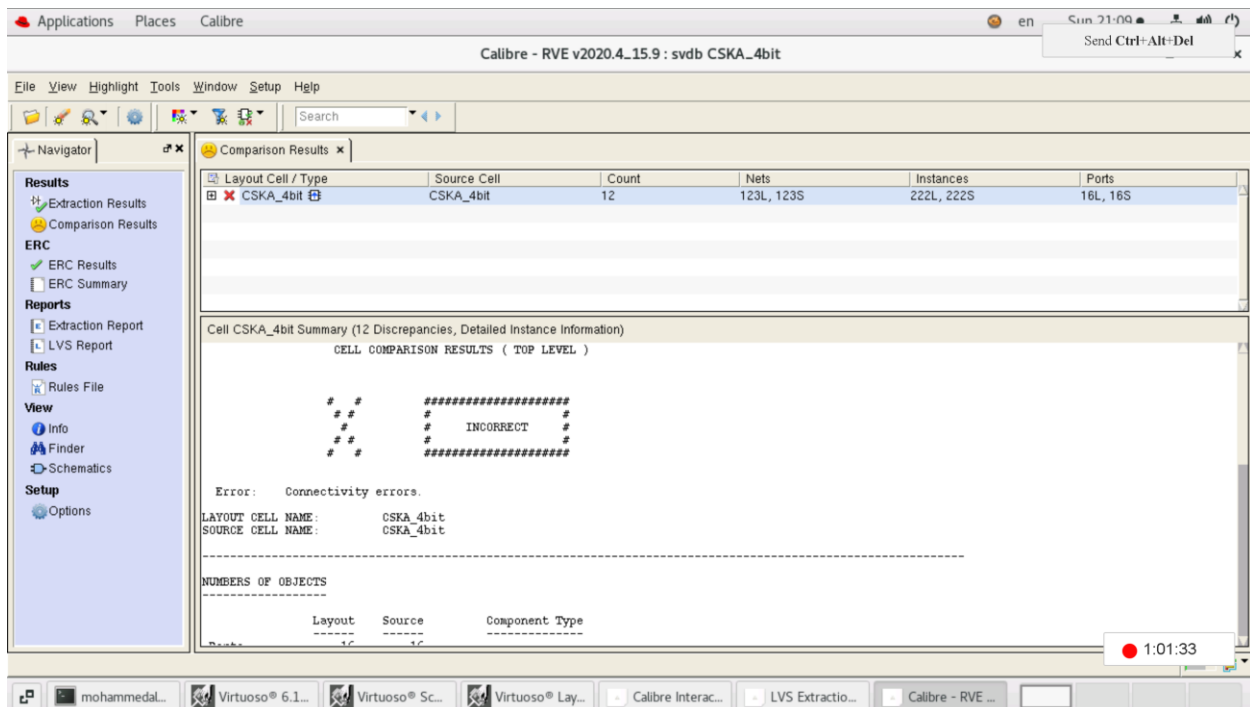


Fig.5 CSKA_4bit LVS

During the LVS verification of the CSKA 4-bit layout, connectivity errors were reported despite the schematic and layout having matching numbers of devices, nets, and ports.

Careful inspection confirmed that all connections were implemented correctly in the layout. Multiple verification and correction methods were attempted, including rechecking contacts, routing, adding new vias, and net labeling; however, the LVS discrepancies could not be resolved. Consequently, although the design was functionally correct at the schematic and simulation levels, a complete LVS match for the CSKA 4-bit layout was not achieved within the project timeframe.

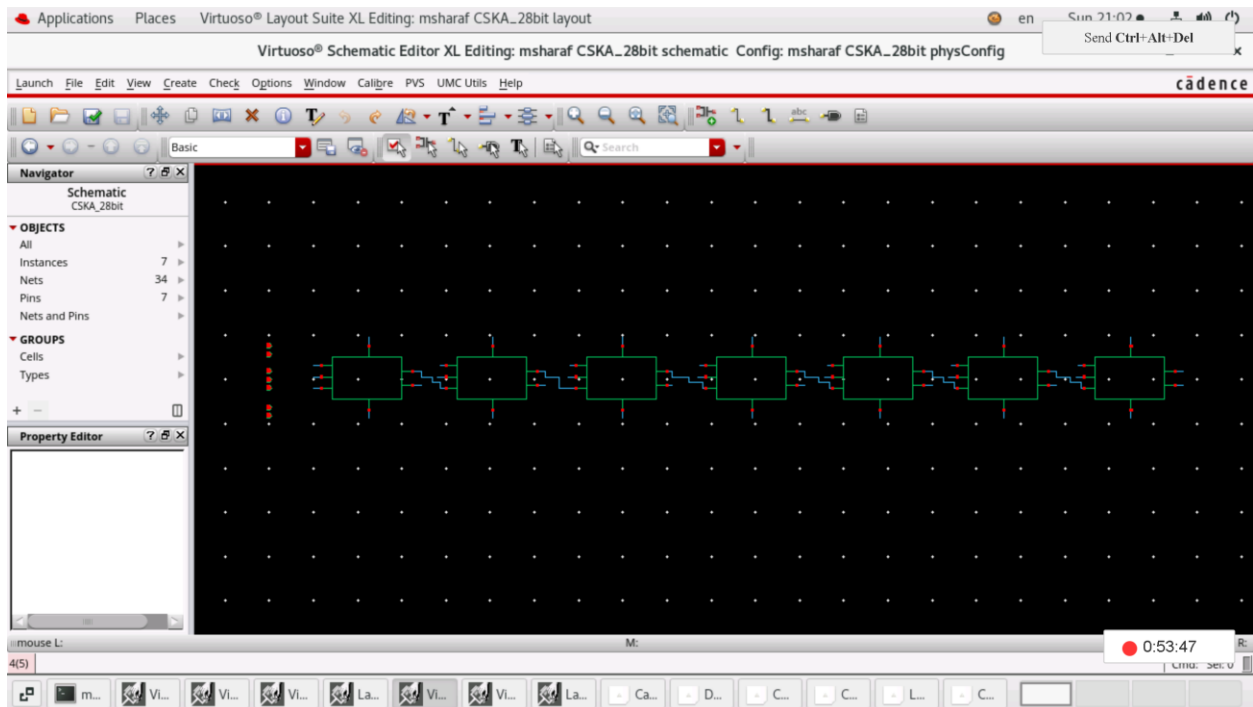


Fig.6 CSKA_28bit Schematic

This schematic illustrates a 28-bit Carry-Skip Adder (CSKA) implemented using a hierarchical structure in Cadence Virtuoso. The design is composed of seven identical 4-bit CSKA blocks connected in series through their carry signals to perform 28-bit addition. Each block generates its local sum outputs while allowing the carry to either ripple through the block or skip ahead when the propagate condition is satisfied. By cascading these CSKA4 blocks, the architecture reduces overall carry propagation delay compared to a purely ripple-based adder while maintaining correct and efficient 28-bit arithmetic operation.

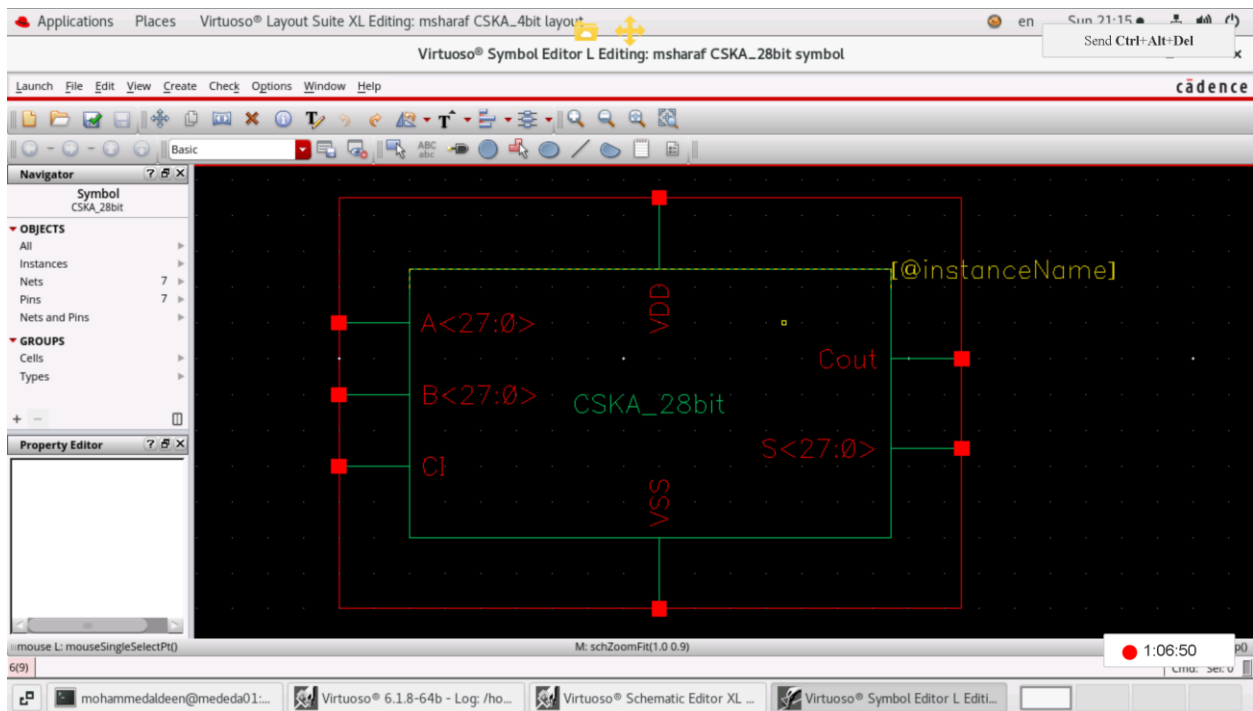


Fig.7 CSKA_28bit Symbol

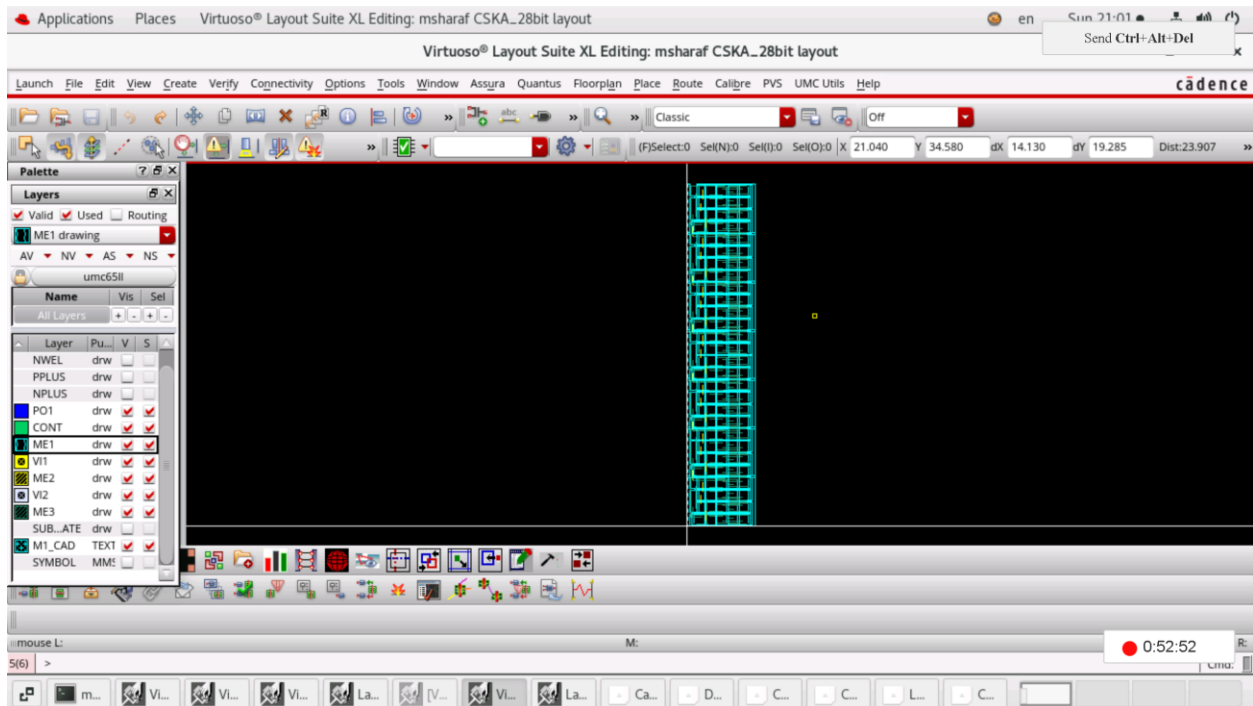


Fig.8 CSKA_28bit Layout

indicated connectivity and net-count mismatches inherited from the lower-level block. Despite this, the DRC checks for both the CSKA 4-bit and CSKA 28-bit layouts were completed successfully with no violations, confirming that the layouts satisfy all physical design rules. The LVS issues therefore stem from schematic–layout connectivity interpretation rather than physical layout errors.

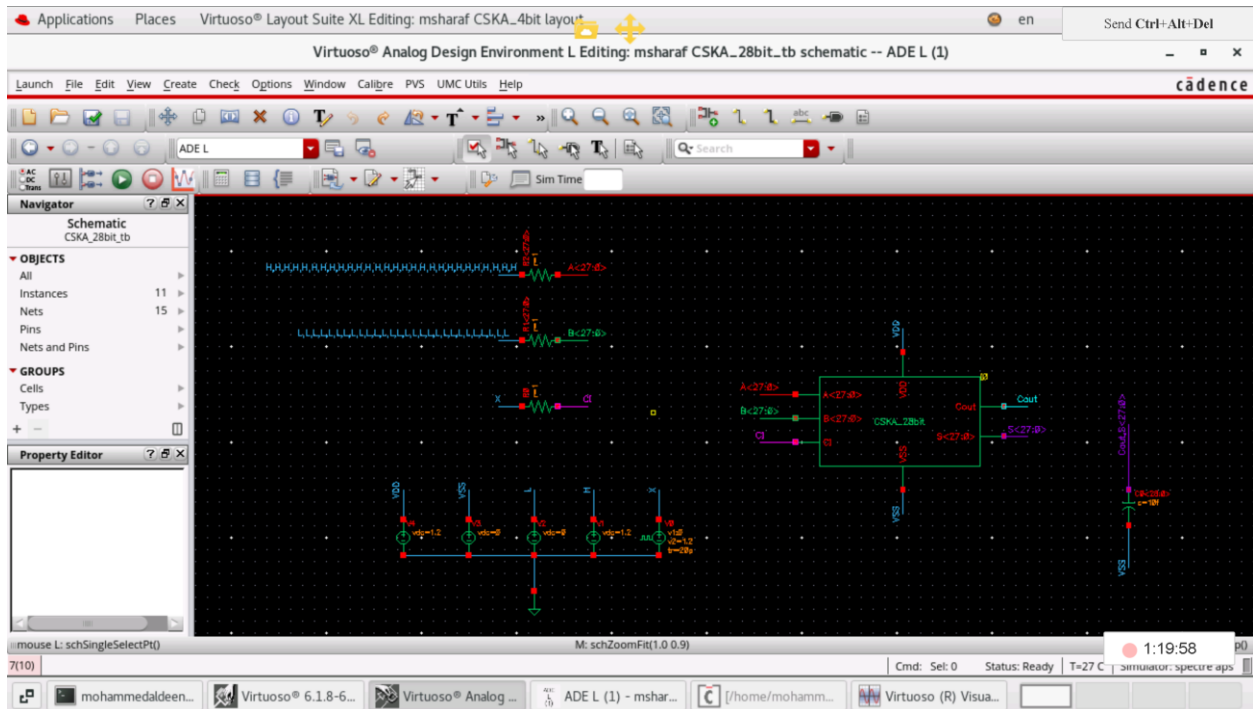


Fig.11 CSKA_28Bits_TB



Fig.12 Fall delay (HL)

- $tp_{HL}=631.055 \text{ ps}-10.0018 \text{ ps}=621.0532 \text{ ps}$



Fig.13 Rise delay (LH)

- $tp_{LH}=2.71572 \text{ ns}-2.03 \text{ ns}=0.68572 \text{ ns}$

The timing performance of the CSA 28-bit adder was evaluated using schematic-level transient simulations. The measured fall propagation delay (HL) was 621.05 ps, and the rise propagation delay (LH) was 0.68572 ns, both well below the 1.2 ns worst-case delay requirement. These results confirm the effectiveness of the carry-skip adder architecture in reducing propagation delay. Post-layout delay analysis could not be performed because the parasitic extraction (PEX) process was not successfully generated; therefore, the reported delays correspond to pre-layout simulations.